Roll Number:_____

***Thapar Institute of Engineering & Technology, Patiala***

Department of Computer Science and Engineering

**MID SEMESTER EXAMINATION**

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| B. E. (Second Year): Semester-IV (2024/25) (2UoQ/2UoT) | Course Code: UCS510<br>Course Name: Computer Architecture and Organization |
| Date: March 5, 2025 | Time: 11:00 a.m. to 1:00 p.m. |
| Duration: 2 Hours, M. Marks: 30 | Name of Faculty: Anju Bala |

***Note:*** **Attempt all questions *in a proper sequence with justification.***
***Assume missing data, if any, suitably.***

<table>
<tr><td>Q1(a)</td><td>Design the 4-bit logic circuit to implement four gates: AND ,NOR, NOT, NAND</td><td>(3)</td></tr>
<tr><td>Q1(b)</td><td>Design an arithmetic circuit for two stages with one selection variable S and data inputs A and B. The circuit generates the following four arithmetic operations in conjunction with the input carry $C_{in}$.<br><table><tr><td>S</td><td>$C_{in}$=0</td><td>$C_{in}$=1</td></tr><tr><td>0</td><td>D=A+B</td><td>D=A+1</td></tr><tr><td>1</td><td>D=A-1</td><td>D=A-B</td></tr></table></td><td>(5)</td></tr>
<tr><td>Q2(a)</td><td>Construct a 5-to-32-line decoder with four 3-to-8-line decoders with enable and one 2-to-4 line decoder</td><td>(3)</td></tr>
<tr><td>Q2(b)</td><td>The value of PC is 9EFHand M[9EF] =F800H, AC=9988H, find the value of PC, IR in Hexadecimal (H) after the execution of INP, OUT, SKI and SKO instructions by taking initial given values.</td><td>(4)</td></tr>
<tr><td>Q3(a)</td><td>Differentiate between direct and indirect addressing mode.</td><td>(2)</td></tr>
<tr><td>Q3(b)</td><td>Show the content in hexadecimal of registers PC, AC, AR, DR, IR and SC of the basic computer. Given the initial content of PC is EFF, the content of AC is 7EC3. The content of memory at address EFF is 8BCD. The content of memory at address BCD is B420. The content of memory at address 420 is FFFF. Give the answer in table with six columns, Repeat the problem with these three memory reference instructions such as ADD, AND and ISZ starting with given initial values.</td><td>(5)</td></tr>
<tr><td>Q4(a)</td><td>Design a combinational circuit with three inputs x, y, z and three outputs A, B, C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When the binary input is 4, 5, 6, or 7, the binary output is one less than the input.</td><td>(5)</td></tr>
<tr><td>Q4(b)</td><td>A computer uses a memory unit with 2K words of 24 bits. A binary instruction code is stored in one word of memory. The instruction has four parts namely a mode bit, an operation code, a register code part to specify one of 60 registers and a signed immediate operand.<br>i. How many bits are there in operation code and register code field?<br>ii. What would be the maximum and minimum value of immediate operand?<br>iii. How many bits are there in the data and address inputs of the memory?</td><td>(3)</td></tr>
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